

Zhiheng Zhang

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Research Profile

Computer Science researcher and research engineer interested in **autonomous program repair, software testing, fault localization, and reliable coding-agent evaluation**. My current systems work focuses on long-horizon execution and on whether agent-generated tests and outputs provide meaningful evidence that intended software behavior has actually been repaired.

Education

University of Melbourne

Master's degree in Computer Science

Melbourne, Australia

Expected 2026

University of British Columbia

Bachelor of Science in Computer Science

Vancouver, Canada

Completed Summer 2024

Selected Research and Engineering Projects

DS-Hns — *Long-horizon autonomous software-work runtime*

2026–present

github.com/zhiheng-zhang-Mera/DS-Hns

- Building an execution layer for unattended software work around the official DeepSeek Harness interface, with optional capabilities isolated as extensions.
- Implemented ordered/scheduled task queues, hardware-adaptive local concurrency, unified task lifecycle and terminal events, task history, and terminal notifications.
- Designed extension-failure isolation, restart/recovery-oriented execution, and an idempotent Windows installer with explicit optional-plugin flow.
- Using the system as a testbed for **reliable coding agents, persistent execution, recovery, scheduling, and completion verification**.

Codex Boss — *Multi-agent orchestration and evidence-based acceptance platform*

2026–present

github.com/zhiheng-zhang-Mera/Codex-Boss

- Building a platform for long-running AI workflows in which multiple agents can plan, execute, critique, and adjudicate work while preserving durable task state.
- Developed platform foundations spanning durable state/events, capability boundaries, plugin isolation, knowledge/data lifecycle, dogfooding, and acceptance/verification.
- Investigating how autonomous systems can distinguish genuine task completion from superficially plausible but weak evidence.
- Current semantic-acceptance work deliberately retains unresolved failure cases: a vacuous case is rejected, while a meaningful case still exposes a false negative in the evidence reader rather than being reported as solved.
- Current regression evidence on the relevant branch includes **2,508 unit tests across 215 files**, plus Electron, renderer, and test typechecking.

Privacy Lens — *Trustworthy-software / privacy project*

- Explored privacy-risk interpretation and user-facing warning mechanisms with attention to bounded interpretation, auditability, evidence boundaries, and trustworthy software behavior.
- Supports broader interests in trustworthy AI, software governance, and high-stakes automated systems.

Quant-Ultra — *Empirical quantitative-research and decision-system project*

2026–present

github.com/zhiheng-zhang-Mera/Quant-ultra

- Developed a personal quantitative-research platform with emphasis on robustness, temporal evaluation, data integrity, risk-aware decision making, and reproducible workflows.
- Shifted the project away from headline-return claims toward evaluation quality, non-stationary data, temporal leakage, walk-forward reasoning, and auditable experiment infrastructure.

Research Interests

Automated program repair; software testing and fault localization; coding-agent validation; regression-aware acceptance; empirical software engineering; long-horizon autonomous software maintenance.

Technical and Research Strengths

- End-to-end research engineering: problem formulation, system design, iterative implementation, testing, debugging, and empirical evaluation.

- Multi-module software design with explicit persistence, lifecycle, extensibility, isolation, recovery, and acceptance criteria.
- Git/GitHub-based experimental development, branch-based iteration, regression testing, and reproducible project documentation.
- Practical multi-AI workflow design: using multiple AI systems as coding/research collaborators while independently reviewing plans, evidence, failure modes, and completion claims.

Current Research Direction

I want to study **repair validation for autonomous coding agents**: how testing and fault-localization signals can distinguish behaviorally meaningful repairs from plausible patches or weak generated tests, and how these methods perform on real repositories over long-running maintenance tasks.